Homework 1: Information Theory for Data Science and Machine Learning

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March 2, 2020

Note: the other tasks or part of them are in the code. The opposite is also true.

1 Problem 1

1.1 a)

Generation of N realization of a random variable X following the univariate Gaussian distribution of mean m and variance v.

1.1.1 Task 1

No precision of whether or not we should create an univarite normal distribution only from the uniform distribution (For instance using Box-Muller transform) or not. So I used directly the **randn** matlab built-in function.

For $N = 1000$, the sample mean, variance and standard deviation most of the time have an absolute error greater than 0.01 compared to their real mean, standard deviation and variance.

1.1.2 Task 2

The absolute errors are lower for $N = 100000$.

1.1.3 Task 3

To explain the above investigation, we can use the Strong Law of Large Numbers (SSLN) for the sample mean. The law explains that the sample mean $\widetilde{X} = \frac{1}{N} \cdot \sum_{k=1}^{N} X_k$ converges almost surely to the expected value:

$$Pr[lim_{N\to\infty}\widetilde{X} = m] = 1$$

with these assumptions:

1. $X_1, ..X_N$ independant
2. Identically distributed

Now we want to prove the convergence of the sample variance (Proving the convergence for the variance implies the convergence for the standard deviation).

We have the sample variance $\frac{1}{N} \cdot \sum_{k=1}^{N}(X_k - m)^2 = \frac{1}{N} \cdot \sum_{k=1}^{N}(X_k^2 - 2 \cdot X_k \cdot m) + m^2 = \frac{1}{N} \cdot \sum_{k=1}^{N} X_k^2 - \frac{2 \cdot m}{N} \cdot \sum_{k=1}^{N} \cdot X_k + m^2 = a - b + c$.

But, b converges to $2 \cdot m^2$ with SLLN, so we have

$$\frac{1}{N} \cdot \sum_{k=1}^{N} X_k^2 - m^2$$

.

But a converges to $E[X^2]$ if we use $Y_k = X_k^2$ we can apply the SSLN, so the sample variance converges to $E[X^2] - E[X]^2 = Var[X]$.

1.2 b)

1.2.1 Generation of the random vector following the multivariate Gaussian distribution

Generation of a random vector $\mathbf{X}$ (one sample/codeword) of dimension N following the multivariate Gaussian distribution of mean vector $\mathbf{m_X}$ and covariance matrix $Cov = \mathbf{K_{XX}}$. To do that, we create a standard normal random vector $\mathbf{Y}$ where $\mathbf{m_Y} = \mathbf{0}$ and $\mathbf{K_{YY}} = \mathbf{Id_N}$ then affine transform it into the random vector we want $\mathbf{X}$ of mean vector $\mathbf{m_X}$ and covariance matrix $Cov = \mathbf{K_{XX}}$.

To do so, we write down $\mathbf{X} = \mathbf{A} \cdot \mathbf{Y} + \mathbf{m_X}$. We can see that we get the correct expectation we wanted from $\mathbf{X}$ which is $E[\mathbf{X}] = \mathbf{A} \cdot E[\mathbf{Y}] + \mathbf{m_X} = \mathbf{m_X}$ because $E[\mathbf{Y}] = \mathbf{0}$.

$$\mathbf{X} = \mathbf{A} \cdot \mathbf{Y} + \mathbf{m_X} \tag{1}$$

Now, we need to know what $\mathbf{A}$ is, to do so we try to get the variance of $\mathbf{X}$ which is $Var[\mathbf{X}] = \mathbf{K_{XX}} = E[(\mathbf{X} - \mathbf{m_X}) \cdot (\mathbf{X} - \mathbf{m_X})^T] = E[(\mathbf{A} \cdot \mathbf{Y}) \cdot (\mathbf{A} \cdot \mathbf{Y})^T] = \mathbf{A} \cdot E[\mathbf{Y} \cdot \mathbf{Y}^T] \cdot \mathbf{A}^T$.

$$Var[\mathbf{X}] = \mathbf{K_{XX}} = \mathbf{A} \cdot E[\mathbf{Y} \cdot \mathbf{Y}^T] \cdot \mathbf{A}^T \tag{2}$$

But $E[\mathbf{Y} \cdot \mathbf{Y}^T] = \mathbf{R_{YY}} = \mathbf{K_{YY}}$ because $Var[\mathbf{Y}] = E[(\mathbf{Y} - \mathbf{m_Y}) \cdot (\mathbf{Y} - \mathbf{m_Y})^T]$ with $\mathbf{m_Y} = 0$.

$$Var[\mathbf{X}] = \mathbf{K_{XX}} = \mathbf{A} \cdot \mathbf{K_{YY}} \cdot \mathbf{A}^T \tag{3}$$

With $\mathbf{R_{YY}} = \mathbf{K_{YY}} = \mathbf{Id_N}$, we have $Var[\mathbf{X}] = \mathbf{K_{XX}} = \mathbf{A} \cdot \mathbf{A}^T$. That means that by using Cholesky decomposition on $\mathbf{K_{XX}}$, we can get $\mathbf{A}$. We can use Cholesky decomposition because the covariance matrix is by definition symmetric and positive definite.

$$Var[\mathbf{X}] = \mathbf{K_{XX}} = \mathbf{A} \cdot \mathbf{A}^T \tag{4}$$

So in summary, we generate a standard normal random vector $\mathbf{Y}$ and apply this formula:

$$\mathbf{X} = \mathbf{A} \cdot \mathbf{Y} + \mathbf{m_X}$$

with $\mathbf{A} \in \mathbb{R}^{N \times N}$ from the Cholesky decomposition. And $\mathbf{X}, \mathbf{Y}, \mathbf{m_X} \in \mathbb{R}^N$.

1.2.2 Marginal distributions

To get the marginal distributions, we can integrate the joint distribution with respect to the other random variables but it's slow in matlab.

Another way is to plot the univariate distributions of each random variable based on the mean vector $\mathbf{m_X}$ and the covariance matrix $\mathbf{K_{XX}}$ like this:

$$f_{X_i}(x_i) = \frac{1}{\sqrt{2 \cdot \pi \cdot \sigma_i^2}} \cdot e^{-\frac{(x_i - m_i)^2}{2 \cdot \sigma_i^2}}$$

with $\sigma_i^2 = \mathbf{K_{XX}}(i,i) = Var[X_i]$ and $m_i = \mathbf{m_X}(i)$ with $i = 1, ..., N$

1.2.3 The different plots from MATLAB

Figure 1: Histogram of Task 1 for the $N = 1000$ realizations following the univariate standard normal distribution

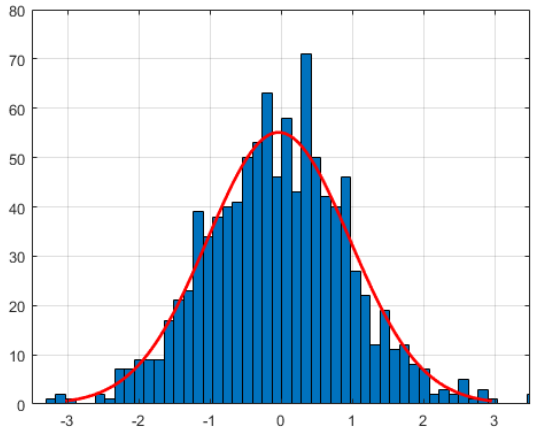


Figure 2: Scatter plot with histogram of $M = 100000$ samples to represent an approximation of the Marginal distributions of Task 2. Note that histograms are not probability density functions..

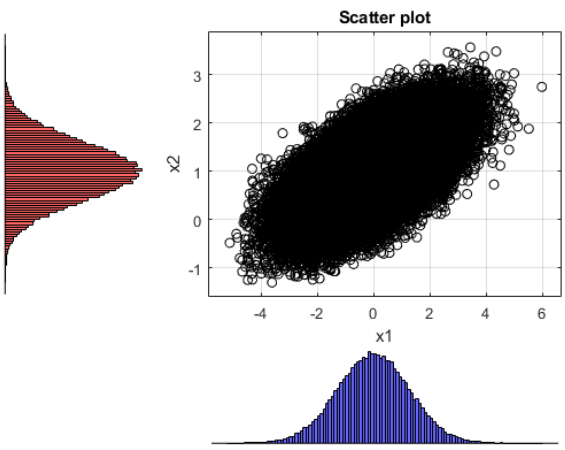


Figure 3: Contour plot of the joint probability density function and plot of the Marginal distributions of Task 2 f_{X1} and f_{X2}

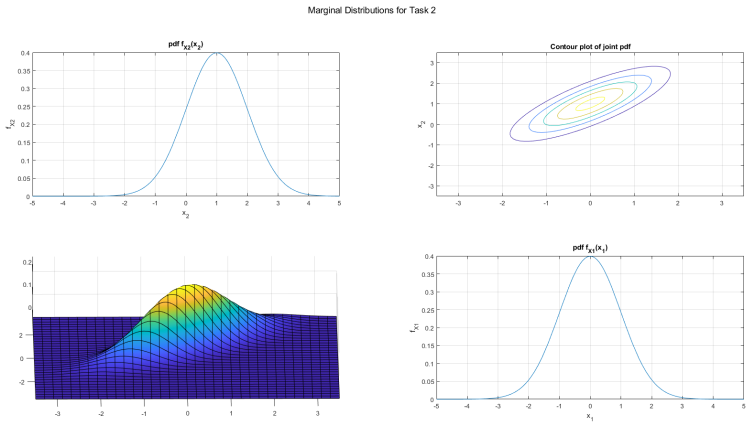
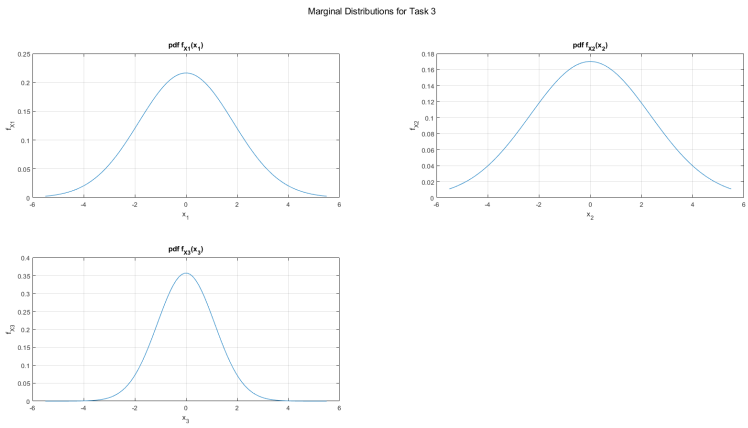


Figure 4: Plot of the Marginal distributions of Task 3 f_{X1}, f_{X2} and f_{X3}



2 Problem 2

2.1 Linear transformations of random vectors

2.1.1 Introduction

A linear transformation has the form $\mathbf{Y} = \mathbf{A} \cdot \mathbf{X}$ while an affine transform has the form $\mathbf{Y} = \mathbf{A} \cdot \mathbf{X} + \mathbf{u}$ with $\mathbf{X}, \mathbf{Y}, \mathbf{u} \in \mathbb{R}^N$ and $\mathbf{A} \in \mathbb{R}^{N \times N}$.

Here, we transform a random vector $\mathbf{z} \sim \mathcal{N}(\mathbf{m_z}, \mathbf{K_{zz}})$ into $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{K_{xx}})$ with $\mathbf{K_{xx}}$ being a diagonal matrix. This transformation is called **decorrelation**.

Affine transform instead of linear transform Supposing $\mathbf{z} \sim \mathcal{N}(\mathbf{m_z}, \mathbf{K_{zz}})$ instead of having the mean vector at $\mathbf{0}$, we want to affine transform this random vector $\mathbf{z}$ into **another one** but with mean vector $\mathbf{0}$ and decorrelated components (for the new random vector). If we write down $\mathbf{x} = \mathbf{A} \cdot \mathbf{z}$, we don't move the mean vector to $\mathbf{0}$, to do so, we want $E[\mathbf{x}] = \mathbf{A} \cdot \mathbf{0}$ so:

$$\mathbf{x} = \mathbf{A} \cdot (\mathbf{z} - \mathbf{m_z}) \tag{5}$$

We can easily verify that $E[\mathbf{x}] = \mathbf{A} \cdot (E[\mathbf{z}] - \mathbf{m_z}) = \mathbf{0}$ and,
$Var[\mathbf{x}] = E[(\mathbf{A} \cdot (\mathbf{z} - \mathbf{m_z})) \cdot (\mathbf{A} \cdot (\mathbf{z} - \mathbf{m_z}))^T] = \mathbf{A} \cdot E[(\mathbf{z} - \mathbf{m_z}) \cdot (\mathbf{z} - \mathbf{m_z})^T] \cdot \mathbf{A}^T$
And $Var[\mathbf{x}] = \mathbf{A} \cdot \mathbf{K_{zz}} \cdot \mathbf{A}^T$

The transformation above move the mean vector to the "center" at $\mathbf{0}$. But to decorrelate the components of the new random vector, we're going to investigate different types of orthogonal matrices $\mathbf{A}$ for random orthonormal transform and show that $\mathbf{A} = \mathbf{U}^T$ with $\mathbf{U}$ from the eigendecomposition is the optimal transform.

2.1.2 Task 1 General linear transform with mean vector at 0

Given $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{K_{zz}})$, with $\mathbf{x} = \mathbf{A} \cdot \mathbf{z}$, we can see as property that the expected value of the random vector $\mathbf{x}$ is $E[\mathbf{x}] = \mathbf{A} \cdot E[\mathbf{z}] = \mathbf{0}$ because to $E[\mathbf{z}] = \mathbf{0}$. The variance using the equation (3) from **1.2.1** under general $\mathbf{A}$:

$$Var[\mathbf{x}] = \mathbf{K_{xx}} = \mathbf{A} \cdot \mathbf{K_{zz}} \cdot \mathbf{A}^T$$

For the intuition, this affine transform squish or stretch space ("linearly") in all the dimensions but keeps the mean vector at $\mathbf{0}$. More precisely, it's a linear transform because we don't move the mean vector. Because it's a general $\mathbf{A}$, we don't know if the random vector $\mathbf{z}$ has uncorrelated components or not. In general, a linear transform squish or stretch space without moving the mean vector, it's only changing the variance.

2.1.3 Task 2 Investigate decorrelation

Investigating **decorrelation** using transform matrix $\mathbf{A}$.

(a) Optimal transform with $\mathbf{A} = \mathbf{U}^T$, and with (3): $\mathbf{K_{xx}} = \mathbf{U}^T \cdot \mathbf{K_{zz}} \cdot \mathbf{U} = \Sigma$

$\mathbf{K_{xx}} = \mathbf{U}^T \cdot \mathbf{K_{zz}} \cdot \mathbf{U} = \mathbf{U}^T \cdot \mathbf{U} \Sigma \mathbf{U}^T \cdot \mathbf{U} = \Sigma$ using the Spectral Decomposition (Eigendecomposition or diagonalization) of $\mathbf{K_{zz}}$.

Σ is a diagonal matrix, so by applying the transformation based on the matrix $\mathbf{A}$, we have with the **new** random vector, **decorrelated components**.

Moreover, the operation is "invertible" using:

$$\mathbf{U} \cdot \Sigma \cdot \mathbf{U}^T = \mathbf{K_{zz}}$$

Or with $\Sigma = \mathbf{K_{xx}}$. So we can after "decorrelation", reconstruct the covariance of $\mathbf{z}$ if the know the matrix $\mathbf{U}$ so going back to the original multivariate normal distribution.

To see why $\mathbf{A} = \mathbf{U}^T$ is better than any orthogonal matrix $\mathbf{A}$, see (b)

(b) Investigation using **random orthonormal transform A** with A being orthonormal : $\mathbf{A} \cdot \mathbf{A}^T = \mathbf{Id_N}$.

With $\mathbf{A}$ orthogonal and $\mathbf{K_{xx}} = \mathbf{A} \cdot \mathbf{K_{zz}} \cdot \mathbf{A}^T$, we can revert from the covariance of $\mathbf{x}$ to the original covariance of $\mathbf{z}$ if we know the matrix $\mathbf{A}$.

However, this matrix $\mathbf{A}$ doesn't always decorrelate the components of the new random vector $\mathbf{x}$. As counter-example, $\mathbf{A} = \mathbf{I_N}$, the resulting multivariate normal distribution doesn't change but for specific matrices like in **Task 2a)** with $\mathbf{A} = \mathbf{U}^T$ from the eigendecomposition, we can decorrelate the components.

2.1.4 Task 3 Examples and plots and intuition

Important To make it easier, we're going to focus on linear transform solely (with mean vector at $\mathbf{0}$). First, we comment the $N = 2$ transformation and show the plots for the transformations using different matrices, then we show the plots for 2-dimensional marginal distributions for $N = 2$ and $N = 4$. Finally we show the scatter histogram (For $N = 2$ only) with 10000 observations.

For $N = 2$, taking the same example as in the lecture notes p.96 with $\mathbf{z}$ as original random vector with covariance matrix and mean vector:

$$\mathbf{K_{zz}} = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} \quad \mathbf{m_z} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

with $\rho = 0.9$

We get as spectral decomposition:

$$\mathbf{U} = \begin{bmatrix} -0.7071 & 0.7071 \\ 0.7071 & 0.7071 \end{bmatrix} \quad \Sigma = \mathbf{K_{xx}} = \begin{bmatrix} 0.1 & 0 \\ 0 & 1.9 \end{bmatrix}$$

We can see that from the new random vector $\mathbf{x}$, the components x_k for $k = 1...N$ are decorrelated because by definition $\mathbf{K_{xx}}(i,j) = Cov(x_i, x_j)$ for $i, j = 1, ...N$ but here for $i \neq j$ $Cov(x_i, x_j) = 0$.

Geometrically/Intuitively (doesn't work if we're trying to compare non linear correlation, for more information search coefficient of correlation in wikipedia), the orientation or "slope" of the cloud of points is related to the linear coefficient of correlation $\rho_{x_i,x_j} = \frac{Cov(x_i,x_j)}{\sigma_{x_i} \cdot \sigma_{x_j}}$ from Pearson so we have :

$$Cov(x_i, x_j) = \rho_{x_i,x_j} \cdot \sigma_{x_i} \cdot \sigma_{x_j}$$

If the "slope" is 0 between x_i and x_j, they're uncorrelated.

Formally, with the definition of the linear coefficient of correlation:

$$\rho_{x_i,x_j} = \frac{Cov(x_i, x_j)}{\sigma_{x_i} \cdot \sigma_{x_j}}$$

We have:

$$Cov(x_i, x_j) = 0 \implies \rho_{x_i,x_j} = 0$$

so x_i and x_j are uncorrelated.

In the following plots, we can see that the $\mathbf{A} = \mathbf{U}^T$ transform is really a decorrelation transform while the $\mathbf{A}$ orthonormal is most of the time producing the density function flipped, sometimes it's decorrelated and sometimes with a different orientation. This transformation is caused by how we generated the matrix.

We created $\mathbf{A}$ orthonormal based of a QR decomposition of a random matrix following an uniform distribution and we also verified that $\mathbf{A}$ is full rank.

For the general $\mathbf{A}$ matrix for linear transform (with the assumptions that the determinant of the random matrix is not 0, otherwise, $det(\mathbf{K_{xx}}) = 0$), we had to verify also that the resulting covariance matrix after transformation is still a symmetric definite positive matrix by using the Cholesky function. We also plot the linear transformation using the close to singular matrices $\mathbf{A}$ to see how bad these transformations can be. A singular matrix is a matrix with determinant 0.

For instance, if the matrix had a 0 as determinant, there's no possible preservation of determinant of the covariances. And note that the covariance matrix and the mean vector completely characterize a multivariate normal distribution.

Recalling the general form for linear transformation:

$$Var[\mathbf{x}] = \mathbf{K_{xx}} = \mathbf{A} \cdot \mathbf{K_{zz}} \cdot \mathbf{A}^T$$

Figure 5: Comparison with different $\mathbf{A}$ with last transformation not "singular"

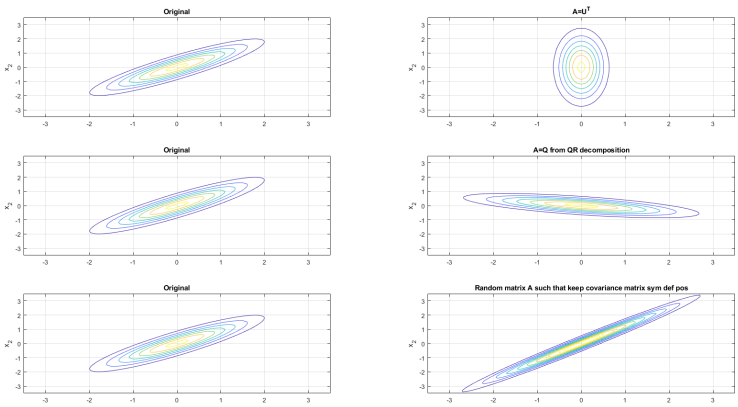


Figure 6: Comparison with different **A** with last transformation not "singular"

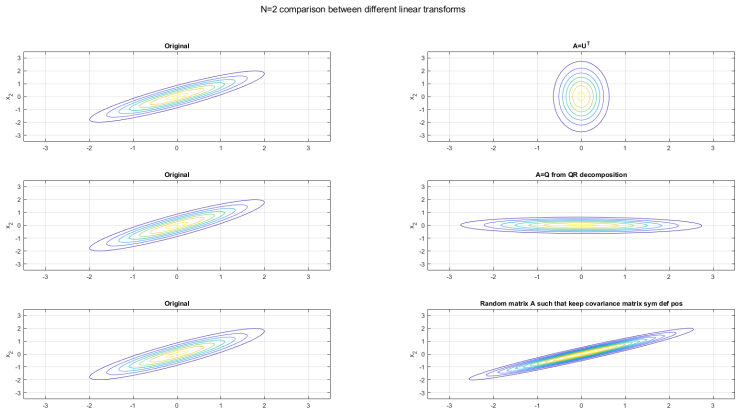


Figure 7: Comparison with different **A** with last transformation close to being "singular"

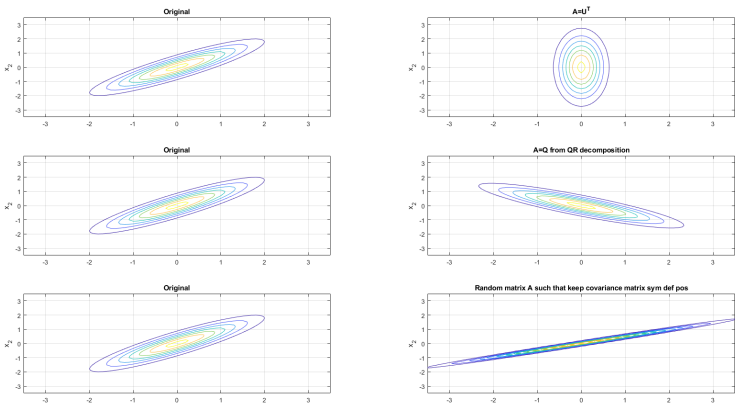


Figure 8: Comparison with different **A** with last transformation really close to being "singular"

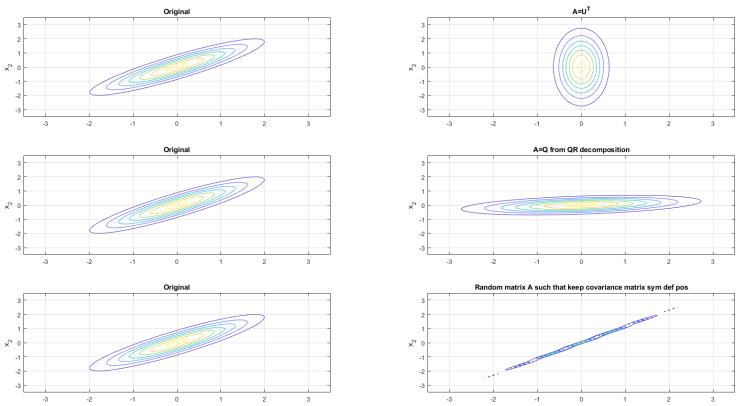


Figure 9: Another random **A** close to being singular

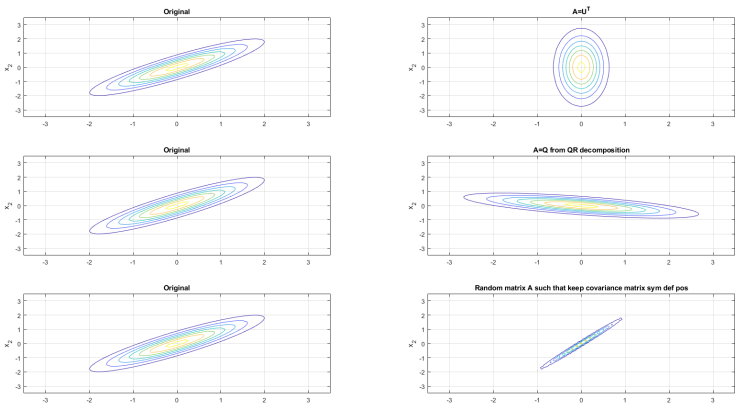


Figure 10: Last one close to being singular we tried

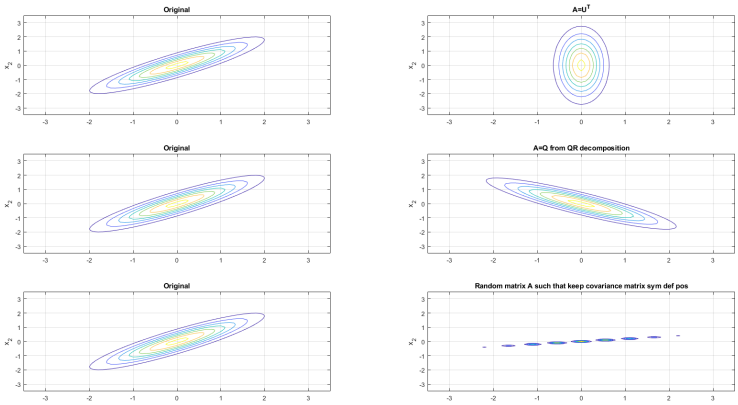


Figure 11: Marginal distributions of $N = 2$ f_{X1} and f_{X2}

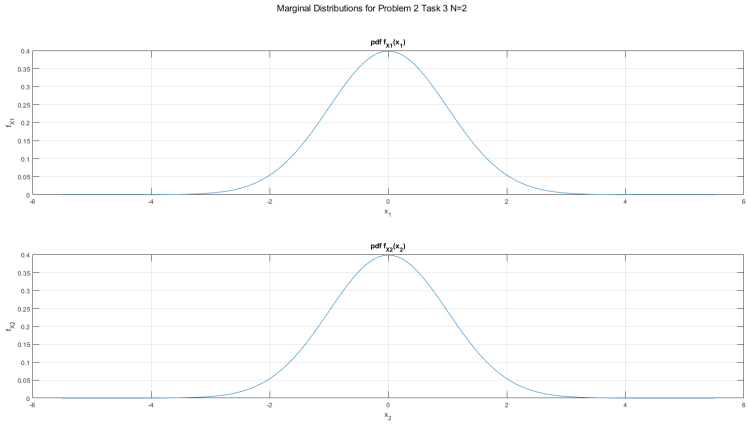


Figure 12: Marginal distributions of $N = 4$ f_{X1}, f_{X2}, f_{X3} and f_{X4}

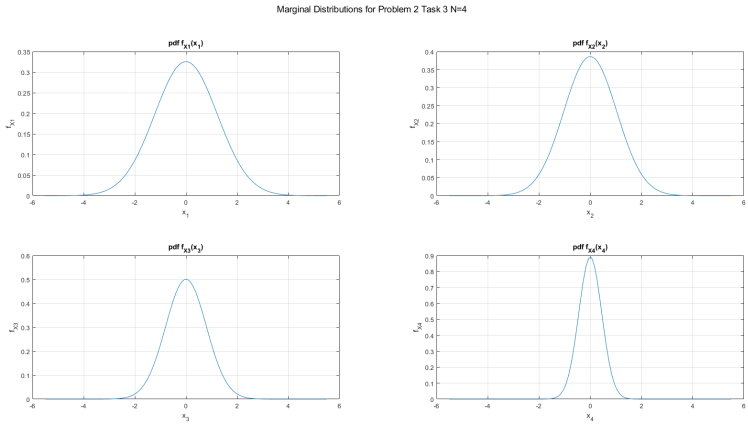
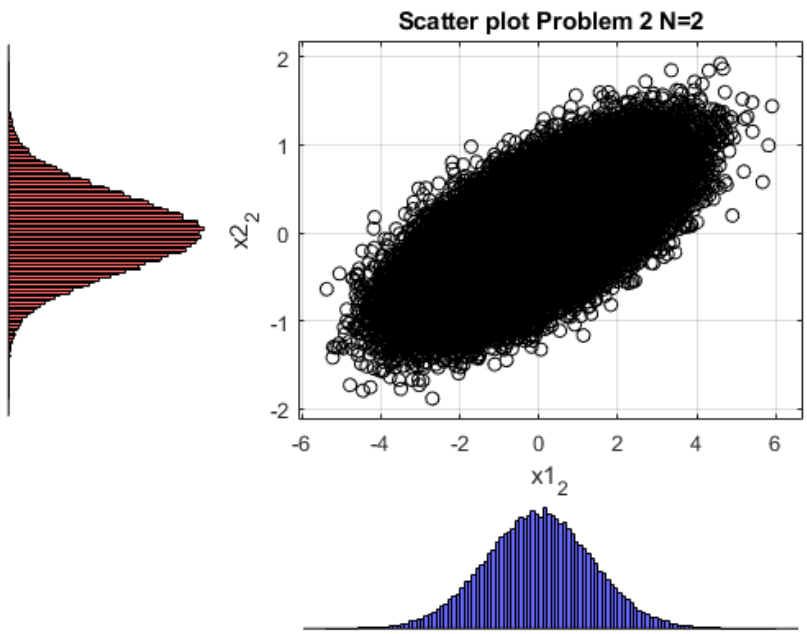


Figure 13: Scatter histogram for the joint pdf defined by $\mathbf{K_{zz}}$ with 10000 observations/samples



2.1.5 Task 4 "Preservation" of determinants, orthonormal transforms

The **determinants don't change**, it means that the covariance is **invariant under the orthonormal transform**. The "volume" corresponding to the application $Var[\mathbf{z}]$ stays the same for application $Var[\mathbf{x}]$.

For linear transform $\mathbf{A} = \mathbf{U}^T$ with $\mathbf{U}$ from the eigendecomposition of $\mathbf{K_{zz}}$

$$det(\mathbf{K_{xx}}) = det(\Sigma) = \prod_{k=1}^{N} \lambda_i \tag{6}$$

$$det(\mathbf{K_{xx}}) = det(\mathbf{U}^T) \cdot det(\mathbf{K_{zz}}) \cdot det(\mathbf{U}) \tag{7}$$

But $1 = det(\mathbf{I_N}) = det(\mathbf{U}^T\mathbf{U}) = det(\mathbf{U}^T) \cdot det(\mathbf{U})$ so using (6):

$$det(\mathbf{K_{xx}}) = det(\mathbf{K_{zz}}) = \prod_{k=1}^{N} \lambda_i \tag{8}$$

For the linear transform where $\mathbf{A}$ is orthonormal in general: $\mathbf{A} \cdot \mathbf{A}^T = \mathbf{I_N}$, we have the same formula as in (7) just with the $\mathbf{U} = \mathbf{A}^T$ so we get

$$det(\mathbf{K_{xx}}) = det(\mathbf{A}) \cdot det(\mathbf{K_{zz}}) \cdot det(\mathbf{A}^T) \tag{9}$$

So we also reach the conclusion that $det(\mathbf{K_{xx}}) = det(\mathbf{K_{zz}})$